

Did Malta's PM2.5 death rate fall by two-thirds?

Testing the EEA's own estimate against its data, its method and Malta's station measurements

THE CLAIM UNDER REVIEW

“... is estimated to have been reduced by 67.7% between 2005 and 2023 (from 143.5 to 46.3, respectively), resulting in 172 (95% CI: 131-192) attributable deaths in 2023.”

European Environment Agency, Air pollution quick country facts (country fact sheets 2025), 1 Dec 2025.

The rate: deaths attributable to PM2.5 above 5 µg/m³ per 100 000 people aged 30+, Malta.

VERDICT: LARGELY SUPPORTED

Accurate report of an estimate; 2005 is a map value

Version 1.0 · 6 October 2026

Status: no right of reply needed

Prepared from public sources and EEA and Eurostat data.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

The European Environment Agency’s *Air pollution quick country facts* (page modified 1 December 2025) say that in Malta the rate of natural deaths attributable to long-term exposure to PM2.5 above 5 µg/m³, per 100 000 people aged 30 or over, “**is estimated to have been reduced by 67.7% between 2005 and 2023 (from 143.5 to 46.3, respectively), resulting in 172 (95% CI: 131-192) attributable deaths in 2023.**”

- 1

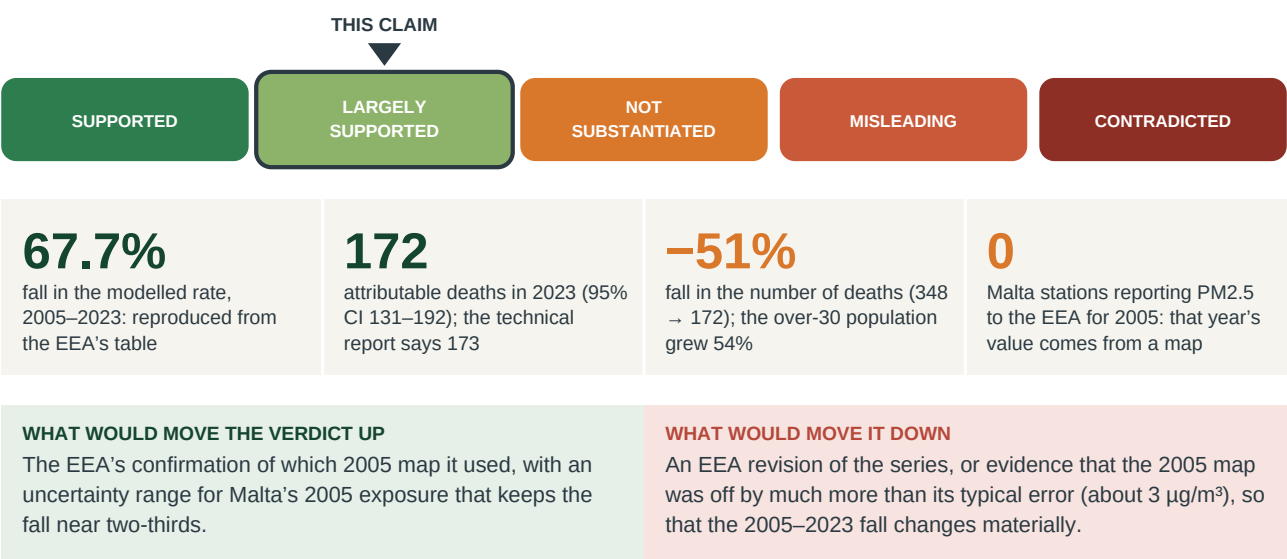
The figures are the EEA’s own, reported exactly. The EEA’s table gives 143.5 (2005) and 46.3 (2023), a fall of 67.74%, and 172 deaths (95% CI 131–192) in 2023; Eurostat republishes the same. The technical report gives 173 (132–193): one death.
- 2

The method is the documented one, and it reproduces. 8% more deaths per 10 µg/m³ (WHO), counted above 5 µg/m³, applied to natural deaths of people aged 30+. From Eurostat’s own death counts we get 346 and 170 deaths for 2005 and 2023, within 1.2% of the EEA’s 348 and 172.
- 3

Measured PM2.5 fell; 2005 is a map value. At Msida PM2.5 fell 39% from 2007 to 2023, and at three long-running stations 25% from 2010: direct evidence of a fall. The EEA’s map is built on these stations, so they anchor it rather than check it. No Maltese PM2.5 was reported to the EEA for 2005; that year’s 21.0 µg/m³ comes from an older map method and model.
- 4

It is a rate, and the start and threshold matter. The number of deaths fell by half (348 to 172) because the over-30 population grew 54%. Counting all concentrations, or starting in 2007, the fall is about 55–57%.
- 5

Verdict: largely supported (high confidence). The statement accurately reports the EEA’s estimate and says it is an estimate. The caveat: the exact 67.7% rests on a 2005 map value, and the 95% CI covers only the risk function.



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3 How the estimate is made	7 Verdict and requests for evidence
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1 The claim and what we could verify

The statement is the EEA’s own text on its *Air pollution quick country facts* page, part of the 2025 air pollution country fact sheets, read in the page HTML on 6 October 2026 [1]. The page gives a sentence of the same form for each country it covers. The EEA is the EU’s environment agency; we assess the wording of its statement, not the agency. Because it is the EEA’s own statistic, the questions are whether the sentence reports the EEA’s estimate accurately, and what that estimate does and does not mean.

What	Where	Access
The Malta sentence (quoted on the cover and in the TL;DR)	Air pollution quick country facts [1]	Read in full, 6 Oct 2026
The estimates by year and scenario, with population and concentration	EEA burden-of-disease table [2]; Eurostat sdg_11_52 [3]	Downloaded, 6 Oct 2026
The method and its history	EEA briefing [4] and indicator [5]; ETC HE technical report [6]; map reports for 2005 [7] and 2023 [8]	[4, 5] read in full; [6–8] pages read
Measured PM2.5 and PM10 in Malta, 2004–2025, and station types	EEA station files [10] and metadata [17]; PQ 29696 table via CC-007 [15]	Downloaded; reused

Our intake record first paraphrased the statement as “PM2.5 deaths down two-thirds”. The EEA speaks of the *rate*, and the number of deaths fell by about half (section 6), so the record title is now “PM2.5 death rate down two-thirds”. The EEA’s *Europe’s environment 2025* page for Malta draws the same series as a chart, with data to 2022 [14]; it is context, not the statement checked.

SCOPE NOTE

This check tests the EEA’s statement about its own estimate. It does not assess Malta’s air-quality policy or any authority, and it does not count deaths: attributable deaths are a statistical estimate for a population, not deaths certified as caused by air pollution (the EEA explains the difference in material linked from its briefing [4]).

2 Method

Question. Do 143.5, 46.3, 67.7% and 172 (131–192) match the EEA’s own estimates; is the method as described; is the 2005 starting point comparable with 2023; and did Malta’s measured PM2.5 fall?

Evidence. We downloaded the EEA’s *Burden of disease of air pollution (Countries & NUTS)* table through its Table publisher [2]: Malta for every scenario, all countries, and the EU-27. We downloaded Eurostat sdg_11_52 (the same estimates, republished [3]), Eurostat population, deaths and causes of death [11] and emissions [12], every Malta PM2.5 and PM10 file in the EEA’s station archive [10] and the EEA’s station metadata [17], all on 6 October 2026 with Eurostat’s flags (none on the Malta values used). *tools/cc-035-report/calc.py* recomputes every figure from *data/cc-035/*. We read the EEA’s briefing and indicator [4, 5], the relevant parts of the technical report behind the estimates [6] and of the two map reports [7, 8], and searched Crossref and Europe PMC for peer-reviewed work on the risk function and on Malta’s PM2.5 [9, 13].

Grades. Official estimates and statistics are grade C, as is the meta-analysis behind the risk function (a review); Malta’s source-apportionment study is grade B. **Verdicts** follow the five-point scale in Appendix A.

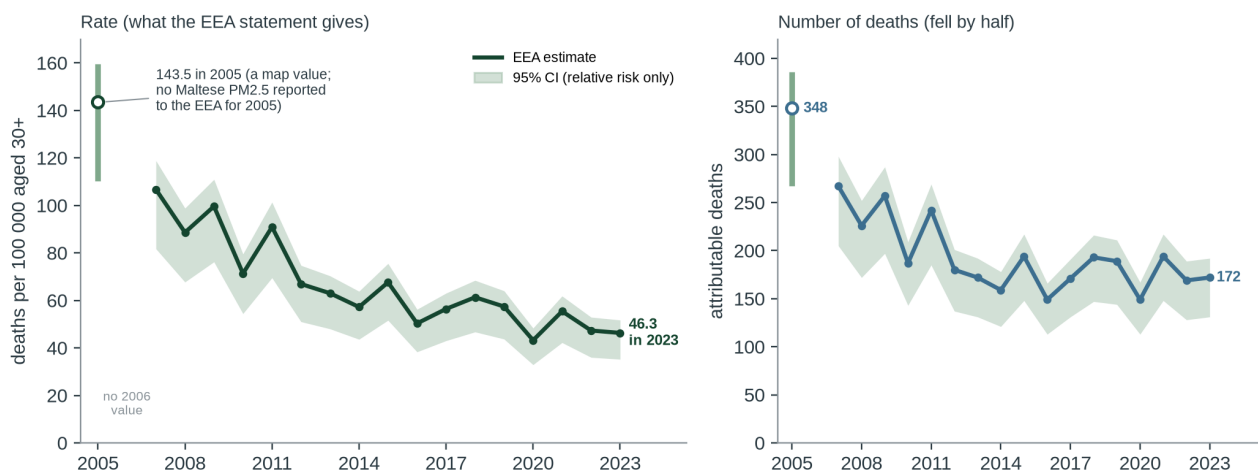
3 How the estimate is made

The EEA's estimate combines four things, cell by cell on a 1 km grid, and adds up the cells [6, Annex 1]:

- **Concentration.** An annual PM2.5 map that interpolates between monitoring stations, with “pseudo” PM2.5 values estimated from PM10 stations, a chemical transport model and other data (altitude, meteorology, population) [6, 8]. For Malta in 2023 its population-weighted mean is 10.9 µg/m³ [6].
- **Population.** Gridded population scaled to Eurostat totals; for all-cause deaths, people aged 30 or over (373,208 in Malta in 2023; Eurostat 373,224 [2, 11]).
- **Baseline deaths.** Deaths from natural causes (all causes except injuries and other external causes) by age and sex, from Eurostat. Cause-of-death data begin in 2011, so 2011 stands in for 2005–2010 [6].
- **Risk.** 8% more natural deaths per 10 µg/m³ (95% CI 6–9%), the WHO 2021 recommendation from Chen and Hoek's meta-analysis of 107 studies [6, 9], counted only above 5 µg/m³, the WHO 2021 guideline level [4].

The attributable fraction in a cell is $1 - \exp(-\beta \times (C - 5))$, with $\beta = \ln(1.08)/10$, and the deaths are that fraction of the cell's baseline deaths [6, Annex 1]. Using Malta's national mean concentration and Eurostat's natural deaths aged 30+ [11], we get 346 deaths for 2005 and 170 for 2023, against the EEA's 348 and 172 (–0.6% and –1.2%). The small gap is expected, since the EEA works cell by cell [6, pp. 45–46].

What changed in the method. The health calculation is now the same for every year. Until 2021 the EEA used a risk of 6.2% per 10 µg/m³ counted from 0 µg/m³; from 2022 it uses 8% counted above 5 µg/m³, and since 2024 it gives rates per person aged 30 or over; it recalculated every year back to 2005 this way [5]. The concentration maps are not one method: PM2.5 maps use an updated mapping method only from 2017, 2005 and 2009 exist in old and updated versions, the ETC's own trend uses old-method maps for 2005–2014 [8, pp. 74–77], and the chemical transport model changed from EMEP to CAMS from 2020 [8, p. 108]. Which 2005 version the EEA's table uses is not stated. No 2006 map was prepared, hence no 2006 value [8, p. 76]. The 95% CI reflects the uncertainty in the risk only; “other possible uncertainties (for instance, the one related to exposure assessments) are not quantified” [4].



Deaths from all natural causes, people aged 30+, attributed to PM2.5 above 5 µg/m³ (WHO 2021 guideline), EEA burden-of-disease table, retrieved 6 Oct 2026. The rate fell 67.7% while the population aged 30+ grew 54% (242,534 to 373,208), so the number fell 50.6%. No 2006 value: no 2006 PM2.5 map was prepared (ETC HE Report 2025/5, p. 76).

Figure 1. The EEA's estimate for Malta, with its 95% CI (an error bar for 2005, which stands alone). Left: the rate the statement gives. Right: the number of attributable deaths. The rate fell 67.7%; the number fell 50.6% because more people were aged 30 or over. 2005 is drawn hollow: it is a map value, with no Maltese PM2.5 reported to the EEA for that year [2, 8, 10].

4 What the measurements show

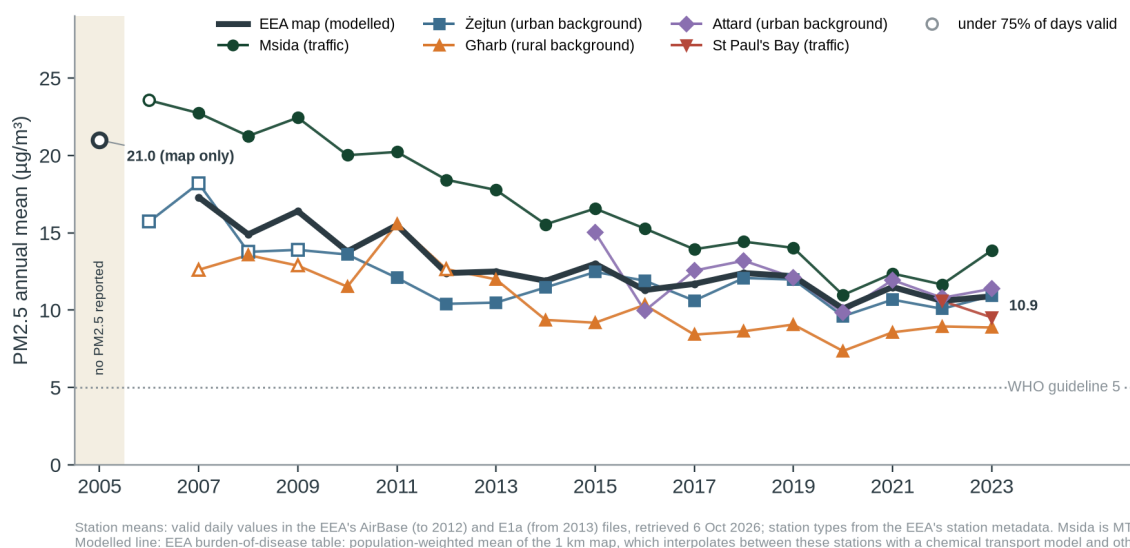


Figure 2. The EEA's modelled population-weighted PM2.5 for Malta (thick line) and the annual means at Malta's stations (thin lines; hollow markers where fewer than 75% of days have valid data), with station types from the EEA's metadata [2, 10, 17]. The map is built on these stations, so it follows them by construction.

Measured PM2.5 fell. At Msida (a traffic station) PM2.5 went from 22.7 $\mu\text{g}/\text{m}^3$ in 2007 to 13.9 in 2023 (–39%); the mean of Msida, Żejtun and Għarb, the three stations with at least 75% of days in every year from 2010 (except 2012), fell from 15.1 to 11.2 (–25%) [10]. This is direct evidence that PM2.5 fell. It is not an independent check of the map: the map interpolates between these same stations [6, 8], so it is anchored to them. In 2023 the five stations in the parliamentary table used in CC-007 average 10.9 $\mu\text{g}/\text{m}^3$, the map's value, but that is an unweighted mean of two traffic, two urban background and one rural station [10, 15, 17].

Two cautions. PM10 at the same stations did not fall (Msida 42.5 $\mu\text{g}/\text{m}^3$ in 2009 and 43.7 in 2023; Żejtun 29.3 in 2007 and 28.7 in 2023) while their PM2.5 fell by about 40% [10]. And the PM2.5 instruments changed: beta-attenuation monitors from 2006, gravimetric reference samplers from 2014, an optical (GRIMM) monitor at Msida from 24 September 2021, and a TEOM-FDMS at Għarb [17]; where two sampling points report the same day, our means average them. We have not untangled these effects.

2005. The EEA's station archive has no Maltese PM2.5 before 29 August 2006 (Żejtun) [10]. For 2005 Malta reported PM10 at two stations: Kordin, which the EEA's metadata classes as industrial, and MT00002, whose type is not in the current metadata [10, 17]. The ETC maps use background and traffic stations only [7, Table A.1], so Kordin probably did not feed the 2005 map, and the 2005 value may rest on no Maltese measurement at all. Our identification of the 2005 map is ETC/ATNI Report 2020/1 (updated method), which gives Malta 21.2 $\mu\text{g}/\text{m}^3$ [7]; the EEA's table has 21.0 [2], a gap we cannot explain. That map's typical cross-validation error was 2.9 $\mu\text{g}/\text{m}^3$ in urban background areas [7]. As a rough test, the 2007 PM2.5/PM10 ratios at Msida (0.54) and Żejtun (0.62) applied to the 2005 PM10 means (49.0 at MT00002, 42.4 at Kordin) give 23–30 $\mu\text{g}/\text{m}^3$ at those two sites, not below the map's 21.0. This is indicative only: one site is industrial, the other's type is unknown, and the ratios come from other sites and another year.

Context. Malta's reported primary PM2.5 emissions fell from 750 to 310 tonnes between 2005 and 2023, with public electricity and heat production down from 420 to 3 tonnes [12]. Emissions are not the whole story: at Msida in 2016, Saharan dust and sea salt made up about a third of PM2.5 [13], and the method treats all PM2.5 mass alike [6, Annex 1].

Indicator	Malta	Comparison	Source	Grade
Rate per 100 000 aged 30+, 2005 → 2023	143.5 → 46.3	EU-27 153.0 → 59.7	[2]	C
Fall in the rate, 2005–2023	67.7%	EU-27 61.0%; Malta 14th of 27	calculated	C
Attributable deaths, 2005 → 2023	348 → 172	EU-27 423,593 → 182,399	[2, 3]	C
Modelled population-weighted PM2.5 (µg/m³)	21.0 → 10.9	EU-27 19.4 → 10.2	[2]	C
Measured PM2.5 at Msida, 2007 → 2023 (µg/m³)	22.7 → 13.9	three stations, 2010 → 2023: 15.1 → 11.2	[10]	C
Natural deaths aged 30+, baseline rate per 100 000	1,238 → 1,038	implied by the EEA figures; Eurostat 1,231 → 1,026	calculated [11]	C

All values in data/cc-035/checks.csv, station_vs_model.csv and eu27_rate_change.csv.

5

Where the evidence points different ways

Q1 Is the 2005 starting point comparable with 2023?		SAME HEALTH METHOD; MAPS DIFFER; 2005 A MAP VALUE
EVIDENCE FOR THE CLAIM'S IMPRESSION The health calculation is the same for every year: the EEA recalculated 2005–2023 with the current risk function, threshold and age group [5, 6]. Our approximation, using the EEA's concentrations and Eurostat's death counts, matches both ends within 1.2%. Measured PM2.5 at Malta's stations fell from 2007 (section 4).	EVIDENCE AGAINST The maps are not one method: PM2.5 maps use the updated method only from 2017, 2005 exists in old and updated versions, and the model changed from EMEP to CAMS from 2020 [8, pp. 74–77, 108]; which 2005 version the table uses is not stated. No Maltese PM2.5 was reported to the EEA for 2005 [10], and both the 2005 and 2023 maps use PM10-based pseudo stations [7; 8, pp. 108, 118]. Shifting Malta's 2005 value by the map's typical error (2.9 µg/m³) gives a fall of 61% to 72% (indicative). The 95% CI leaves exposure error out [4].	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: the health side is consistent; the exposure side is not one method, and the 2005 value is a map value. The fall is less certain than its one decimal place suggests. That PM2.5 fell is borne out by measurements from 2007.		
Q2 Does a 67.7% fall in the rate mean two-thirds fewer deaths?		ACCURATE AS WORDED
EVIDENCE FOR THE CLAIM'S IMPRESSION The EEA says “rate”, names the threshold (above 5 µg/m³) and the age group, and says “is estimated”. A rate per person at risk is how the EEA compares countries [4]. Eurostat's rate per inhabitant of all ages fell 64% [3].	EVIDENCE AGAINST The number of deaths fell 50.6% (348 to 172), because the over-30 population grew 54% [2]. Part of the rate's fall is a 16% lower baseline death rate among people over 30, not cleaner air (section 6). Counting all concentrations the fall is 54.7% [2]. And 5 µg/m³ is not a no-effect level: studies at low concentrations find similar or higher risks [9], and the EEA notes no evidence of a threshold [4].	
WHY THEY DIFFER, AND WHAT IT MEANS HERE For this claim: the sentence answers one well-defined question correctly. It does not say that two-thirds fewer people died from PM2.5, and readers should not take it that way.		

6

Testing the claim

Sub-claim	What the evidence shows	Rating
A. The rate was 143.5 per 100 000 aged 30+ in 2005	EEA table: 143.5 (348 deaths, 242,534 people aged 30+) [2]; Eurostat: 348 deaths [3].	ACCURATE
B. It was 46.3 in 2023	EEA table and technical report: 46.3 (35.2–51.7) [2, 6].	ACCURATE
C. “reduced by 67.7% between 2005 and 2023”	$(143.5 - 46.3) / 143.5 = 67.74\%$. The 2005 value is a map value, from maps of another method and model, with no Maltese PM2.5 reported to the EEA [8, 10]; from 2007 the fall is 56.6%; the 95% CI covers the risk only [4].	LARGELY ACCURATE
D. “172 (95% CI: 131-192) attributable deaths in 2023”	EEA table and Eurostat: 172 (131–192) [2, 3]. Technical report: 173 (132–193) [6]. The rate implies 172.8 deaths, but rounding does not explain the lower bound (35.2 → 131.4): a one-death difference, unexplained.	ACCURATE
E. Natural deaths, long-term exposure above 5 µg/m³, people aged 30+	Matches the documented method [4–6]; recomputed from Eurostat deaths within 1.2% (section 3).	ACCURATE

Why the rate fell. Splitting the change: the attributable fraction fell from 11.6% to 4.4% of natural deaths ($\times 0.383$), as the modelled excess above 5 µg/m³ fell from 16.0 to 5.9 µg/m³; and the baseline natural death rate among people aged 30+ fell 16% ($\times 0.838$) [2, 11]. Together: $\times 0.321$, a fall of 68%. On the attributable fraction alone the fall would be 61.7%.

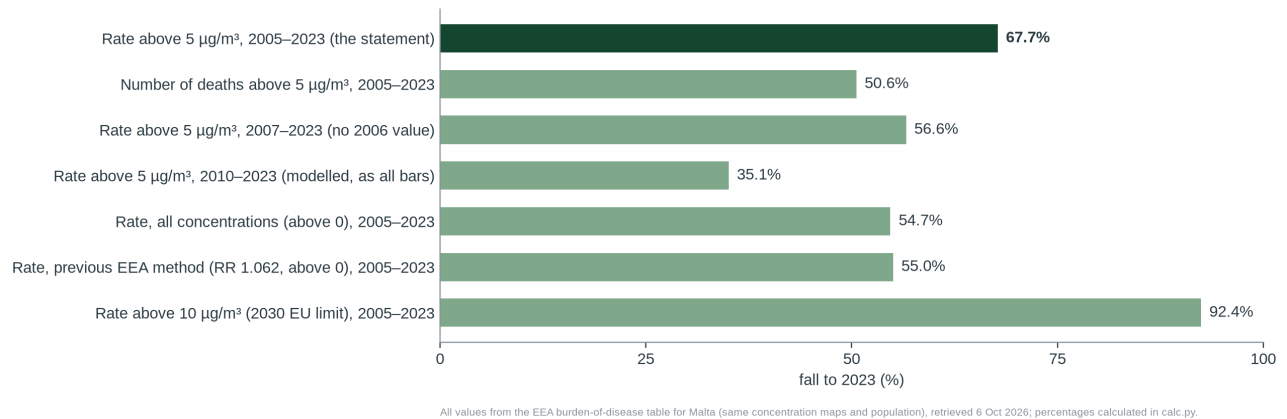


Figure 3. The same EEA estimates measured different ways; every bar is the EEA’s modelled estimate. Only the top bar is the statement; the others show how the size of the fall depends on the threshold (10 µg/m³ is the 2030 EU limit value [8, p. 76]), the method, the start year and whether rates or numbers are compared [2].

7 Verdict and requests for evidence

VERDICT

LARGELY SUPPORTED

The statement reports the EEA's own estimate exactly and calls it an estimate; its 2005 starting value is a map value, with no Maltese PM2.5 reported to the EEA for that year. Confidence: high.

Why. (1) Every figure in the sentence reproduces exactly from the EEA's own table (Eurostat republishes the same estimates, so it is not a separate source). (2) Recomputed from Eurostat's own death counts and the EEA's concentrations, the deaths come within 1.2% of the EEA's, so the method is as described. (3) The sentence is carefully worded: a rate, above 5 µg/m³, people aged 30+, "is estimated". (4) Measured PM2.5 fell at Malta's stations from 2007, direct evidence that the exposure behind the estimate fell; the map is built on these stations, so their agreement with it is not a separate check. (5) The caveat is the start: the 2005 value is a map value, from an older map method and model, with no Maltese PM2.5 reported to the EEA, and the 95% CI leaves exposure error out, so the exact 67.7% is less certain than it looks. That is a minor caveat; it does not change the substance, a large fall. Confidence is high because three separate lines agree: the figures reproduce exactly, the deaths reproduce from Eurostat's death counts, and measured PM2.5 has declined since 2007. We do not count Eurostat's `sdg_11_52` (the EEA's estimate republished) or the station–map match (the map is anchored to the stations).

What this verdict does not say. It does not say that two-thirds fewer people died because of PM2.5 (the number fell by about half), that Malta's air meets the WHO guideline (10.9 µg/m³ against 5 in 2023), or anything about the causes of the fall or any authority's policy.

Evidence we are asking for

- ☐ From the EEA: which 2005 PM2.5 map (old or updated method) the burden-of-disease table uses, and an uncertainty range for Malta's modelled exposure in 2005, which the 95% CI does not include.
- ☐ From the EEA: why the quick-facts page gives 172 (131–192) and the technical report 173 (132–193).
- ☐ From the Environment and Resources Authority, which runs Malta's stations: any 2005 PM2.5 measurements not reported to the EEA, and the type of station MT00002.

8 Limitations

- The statement is a modelled estimate; we tested it against the EEA's own data, its documented method and measured concentrations, not against an independent mortality study for Malta (we found none).
- Our recomputation uses Malta's national mean concentration, not the EEA's 1 km grid, so it is a check on the order of the result, not a replication.
- The stations anchor the map, so they show that PM2.5 fell but cannot test the map's values; the 2023 match (10.9) is largely by construction.
- PM10 at Msida and Żejtun did not fall while PM2.5 did, and the PM2.5 instruments changed over the period (section 4); we have not untangled the two.
- The map errors we quote are Europe-wide cross-validation statistics, not Malta-specific; the 61–72% range and the 2005 PM10-ratio test are indicative.
- The 2005 map is our identification (ETC/ATNI 2020/1); the EEA does not say which version it used.
- Station annual means are averages of valid days from the EEA's files, not the authority's official annual statistics; for 2023 they match the PQ 29696 table to 0.1 µg/m³.

- Chen and Hoek [9] and Scerri et al. [13] were read as abstracts; a corrigendum to Scerri et al. (*Chemosphere* 219:1061–1062, 2019) was not read (Europe PMC lists no abstract and no open-access text).
- The quick-facts and Malta pages were not archived (the Wayback Machine refused requests on 6 October 2026).

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Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand (none in this report).

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	6 Oct 2026	First issue. Verdict Largely supported (high confidence); no right of reply needed.